

Notes: Lerner (JF, 1995)

Venture Capitalists and the Oversight of Private Firms

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1 Big picture

- **Question.** Do venture capitalists act as intensive monitors of the private firms they finance? More concretely, is their formal board representation stronger when the need for oversight is greater, and when the cost of oversight is lower?
- **Setting.** The paper studies **U.S. private biotechnology firms** financed by venture capital between **1978 and 1989**. It combines corrected Venture Economics financing data with board information from SEC filings, industry directories, and the North Carolina Biotechnology Center.
- **Core challenge.** A venture capitalist board seat may reflect monitoring, but it could also reflect ownership, certification, investor prestige, or simple board expansion as firms raise more money. Geographic proximity could matter because of local deal sourcing or local networks, not only because monitoring is costly.
- **Main conceptual contribution.** The paper turns *board membership* into an observable proxy for venture capitalist oversight. It then uses two comparative statics to test whether the pattern looks like monitoring:
 - board representation should rise when monitoring needs rise, and
 - board representation should be more likely when monitoring is cheaper to provide.
- **Main hypotheses.**
 - **Oversight-need hypothesis:** around **CEO turnover**, firms face organizational disruption and uncertainty about the new chief executive, so monitoring demand should rise.
 - **Monitoring-cost hypothesis:** because effective oversight involves visits, meetings, and active involvement, venture capitalists should be more likely to sit on the boards of **nearby** portfolio firms.
- **Main empirical design.** The paper first compares changes in board composition between financing rounds with and without CEO turnover. It then estimates:
 - a **Poisson regression** for the number of newly added directors by type, and
 - a **probit regression** for whether a given venture investor serves on the board, as a function of geographic distance and investor characteristics.
- **Main baseline result.** Between financing rounds with no CEO turnover, firms add only about **0.24** venture directors on average. When CEO turnover occurs, they add about **1.75** venture directors. The increase in **other outsiders** is essentially unchanged, at **0.28** versus **0.33**.
- **Key regression result.** In the Poisson specification, the coefficient on CEO turnover in the venture-capitalist addition regression is **1.88** with a t-statistic of **8.86**. At the mean of the controls, CEO turnover raises the expected number of new venture directors from about **0.25** to **1.59**. The corresponding coefficient for other outsiders is **-0.04** and insignificant.
- **Key board-structure fact.** Boards grow as financing rounds progress. Mean board size rises from about **4.06** members in first rounds to about **5.66** in fourth and later rounds, with

venture capitalists holding about **2.12** seats in later rounds. That is very close to Barry et al.'s inter-industry IPO benchmark of roughly **two of six** seats.

- **Key proximity result.** Venture directors tend to be local. The nearest venture director has mean distance **359 miles**, median **59 miles**, and first quartile **7 miles**. A venture investor with an office within **5 miles** of the firm serves on the board with probability **46.7%**; if the nearest office is more than **500 miles** away, the probability is only **21.8%**.
- **Main credibility result.** The CEO-turnover effect is specific to venture capitalists rather than outsiders in general, and the distance result survives controls for equity stake, venture capitalist age, and venture capitalist size.
- **Economic takeaway.** Venture capitalists appear to be *active, local, state-contingent monitors*, not passive providers of funds. Their governance role becomes especially visible when firms are under stress and when direct oversight is easier to deliver.

2 Data and measurement (key objects)

2.1 Why biotechnology is the right laboratory

- Biotechnology firms are a useful setting because they are young, high-uncertainty firms with few tangible assets, exactly the sort of companies that banks find hard to finance and that venture capitalists are thought to monitor closely.
- The single-industry focus gives the paper unusually rich information from specialized sources, allowing the author to track firms that later **went public**, were **acquired**, or were **terminated** before going public.
- The paper reports that the board structure in this biotech sample looks very similar to the broader inter-industry population of venture-backed IPOs in Barry et al. (1990), so the industry is special enough for measurement but not obviously bizarre in governance structure.

Interpretation. The biotech focus improves data quality and comparability, while still speaking to a broader question about venture-capital monitoring.

2.2 Data source and sample construction

- The starting point is the Venture Economics database of private venture financings.
- The database initially identifies **307** biotechnology firms that received venture capital as private firms between **1978 and 1989**.
- The author removes:
 - **13** foreign firms financed by U.S. providers,
 - **4** buyouts or divisional spin-outs,
 - **3** duplicate entries under different names,

- and **16** firms that received venture capital only after going public.
- This leaves a core sample of **271 private biotechnology firms**.

The author then looks for omitted firms using SEC filings, Recombinant Capital records, industry directories, press databases, and direct contacts with firms and venture capitalists. This search uncovers **37 additional U.S. biotech firms** that had received venture capital while private.

Interpretation. The paper does not simply trust a commercial venture database; it performs an explicit coverage and correction exercise.

2.3 How complete is the sample?

The paper evaluates the importance of the 37 omitted firms using three external benchmarks:

- the sample accounts for over **98%** of biotechnology patent awards to venture-backed firms,
- over **95%** of biotechnology “actions” database entries,
- and over **91%** of external financing raised by venture-backed firms between 1978 and 1989.

Interpretation. The omitted firms exist, but they appear to be economically smaller and less central than those retained in the main sample.

2.4 Corrected financing-round sample

- After corrections, the final financing sample contains **653 financing rounds** for **271 firms**.
- Financing rounds are concentrated in the later part of the 1978–1989 period.
- The reporting of total external financing amounts in Venture Economics is roughly unbiased: the ratio of reported to actual financing is about **1.04**.
- The *number* of rounds is overstated: the commercial database reports about **28%** more rounds than actually occurred.
- The spurious extra rounds are more common in older firms and earlier records, likely because one round is sometimes split into multiple records when cash is disbursed in stages or sources disagree.

Interpretation. The financing amounts are usable, but round counts require cleaning. The paper does that cleaning explicitly.

2.5 What Table I says about the financing process

Table I shows the staged nature of venture financing.

- First rounds: **270** rounds with mean size **\$2.11 million**.
- Second rounds: **186** rounds with mean size **\$3.94 million**.

- Third rounds: **113** rounds with mean size **\$6.14 million**.
- Fourth and later rounds: **84** rounds with mean size **\$5.94 million**.

Interpretation. As in standard venture-capital staging logic, later rounds are fewer but substantially larger.

2.6 Board-membership data and director categories

Board information is collected from:

- IPO prospectuses and other SEC filings,
- filings related to acquired firms and aborted IPOs,
- and files from the North Carolina Biotechnology Center.

The paper classifies directors into four groups:

- **Venture capitalists:** general partners or associates at venture organizations, including full-time affiliates placed in portfolio firms.
- **Other outsiders:** corporate investors, other significant investors, and disinterested outsiders.
- **Insiders:** senior and junior managers directly employed by the firm.
- **Quasi insiders:** parties with an ongoing relation to the firm but not direct employment, such as affiliated academics, lawyers, bankers, former managers, or relatives.

Interpretation. The key distinction is between venture capitalists and *other* outsiders. The main test is whether CEO turnover brings in specifically venture capitalists, not just any outsider.

2.7 Defining CEO turnover

The paper counts only cases where the firm's *top executive* is replaced. It excludes cases that would create spurious turnover:

- initial hiring of a first real CEO,
- title reshuffling in which a venture capitalist gives up "chairman and CEO" but remains involved,
- and acting-CEO arrangements replaced by a full-time chief.

The final CEO-turnover sample contains **40** cases.

- The median age of exiting CEOs is only **40**.
- Only **one** replaced CEO is between ages 64 and 66.

Interpretation. These are not mostly retirement events. The paper treats them as episodes of organizational stress and heightened monitoring need.

2.8 Defining geographic proximity

For each venture organization, the author identifies the office *nearest* to the firm’s headquarters and computes the mileage between the centers of the two ZIP codes.

For the proximity regressions, the unit of observation is a **venture investor–firm pair** measured at the time of the **last venture round** in the sample.

The board-membership dependent variable is:

$$Board_{vf} = 1 \tag{1}$$

if a representative of venture organization v serves on firm f ’s board at the last observed round, and zero otherwise.

Interpretation. The distance measure is not from the VC partnership’s headquarters in general; it is from the *nearest office actually available* to oversee that firm.

2.9 Dependent variables

The paper studies two kinds of outcomes.

First, for financing-round comparisons:

- the change in the number of venture-capitalist directors,
- the change in the number of other outsiders,
- and, descriptively, changes in insiders and quasi insiders.

Second, for proximity tests:

- whether a given venture investor serves on the board at the last round.

2.10 Descriptive facts about boards

Table II provides the main board facts.

- Mean board size rises from about **4.06** in the first round to about **5.66** in later rounds.
- Venture-capitalist seats rise from **1.40** in first rounds to **2.12** in later rounds.
- Other outsiders rise from **0.86** to **1.27**.
- Insiders rise from **1.28** to **1.73**.
- Quasi insiders rise from **0.52** to **0.67** before falling slightly to **0.54** in later rounds.

At the last financing round, the composition of all board seats is:

- venture capitalists: **36.2%**,
- senior managers: **20.3%**,

- affiliated academics: **8.9%**,
- junior managers: **7.1%**,
- corporate partners: **6.4%**,
- with the remaining share spread across other investors, outside executives, outside professionals, and other quasi insiders.

Interpretation. Venture capitalists are the single largest identifiable governance bloc on these private-firm boards.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework and hypotheses

The paper follows Fama and Jensen (1983) and Williamson (1983): when the risk of managerial deviation is higher, boards should take on more monitoring responsibility and outsiders should have greater representation.

Lerner’s additional step is to treat venture capitalists as a particularly important subset of outsiders because they are informed investors with intensive involvement in portfolio firms.

Two testable implications follow:

- board representation by venture capitalists should increase when oversight need rises;
- board representation by venture capitalists should be more likely when oversight is cheaper to provide.

3.2 Event-style comparison around CEO turnover

The simplest empirical design compares average changes in board seats between financing rounds with and without CEO turnover:

$$\Delta Directors_r^{type} = E[\Delta Directors_r^{type} \mid CEO Turnover_r = 1] - E[\Delta Directors_r^{type} \mid CEO Turnover_r = 0]. \quad (2)$$

The paper computes this separately for:

- venture capitalists,
- other outsiders,
- insiders,
- quasi insiders.

Interpretation. This is not yet a full regression design. It is the direct “does board composition shift differently when the CEO changes?” comparison.

3.3 Poisson regression for newly added directors

The core regression takes the count of newly added directors as the dependent variable and estimates separate equations for venture capitalists and other outsiders:

$$E[NewDir_{s_r}^{type} | X_r] = \exp(\alpha + \beta CEO_{Turnover}_r + \gamma \Delta Funding_r + \delta Departures_r). \quad (3)$$

where:

- $CEO_{Turnover}_r$ indicates whether the CEO changed between the previous and current venture round,
- $\Delta Funding_r$ is the difference in dollars invested between the two rounds,
- $Departures_r$ is the number of directors who left the board between the two rounds.

The paper uses all **216 second and later venture rounds** for which both board composition and financing amounts are known at the current and previous rounds.

Interpretation.

- β asks whether CEO turnover predicts more new directors of a given type.
- γ controls for the possibility that bigger financing rounds bring in new investors who naturally demand board seats.
- δ controls for mechanical seat replacement.

3.4 Why a Poisson specification?

The dependent variable is a nonnegative count: the number of newly added directors. Following Hermalin and Weisbach (1988), the paper uses a Poisson model and notes that a goodness-of-fit test does not reject the Poisson specification.

Interpretation. This is a natural reduced-form count model rather than a structural board-choice model.

3.5 Tests of specificity across director types

The paper does not stop at showing that CEO turnover predicts more venture directors. It explicitly tests whether the CEO-turnover coefficient is the same in the venture-capitalist and other-outsider regressions.

In Table IV, Panel B, the χ^2 test rejects equality with a p-value of **0.000**.

Interpretation. The evidence is not “firms add outsiders in general around CEO changes.” It is specifically that they add *venture capitalists*.

3.6 Probit model for board membership and geographic proximity

For investor–firm pairs, the paper estimates:

$$\Pr(Board_{vf} = 1 \mid Z_{vf}) = \Phi(\alpha + \beta Distance_{vf} + \gamma Stake_{vf} + \eta Experience_v + \kappa Size_v), \quad (4)$$

where:

- $Distance_{vf}$ is the distance in **thousands of miles** from the VC’s nearest office to the firm’s headquarters,
- $Stake_{vf}$ is the fraction of the firm’s equity held by the venture organization,
- $Experience_v$ is venture capitalist age,
- $Size_v$ is the organization’s share of the total venture-capital pool.

Because age and size are highly correlated, the paper runs separate specifications using one or the other.

Interpretation.

- $\beta < 0$ means more distant venture investors are less likely to serve on the board.
- $\gamma > 0$ means investors with more skin in the game are more likely to sit on the board.
- Positive coefficients on age or size are consistent with certification, experience, or organizational capacity effects.

3.7 Main coefficient patterns

The key regression results are straightforward.

- In the venture-capitalist Poisson regression, the CEO-turnover coefficient is **1.88 [8.86]**.
- In the other-outsider Poisson regression, the coefficient is **-0.04 [0.13]**.
- In the probit specifications, the distance coefficient ranges from about **-0.16** to **-0.20**, always highly significant.
- The stake coefficient is about **4.19–4.21** and strongly positive.
- Venture-capitalist age is small but positive, about **0.01**.
- Venture-capitalist size, measured as share of the total venture pool, is strongly positive, about **18.6**.

Interpretation. Both monitoring-cost and investor-importance variables matter, but geographic proximity remains robust after controlling for ownership and venture-organization characteristics.

3.8 Weighting, outliers, and sample scope

- The paper does not rely on special weighting schemes in the reported board regressions.

- The board-composition table uses **362 financing rounds** where board membership can be determined.
- The CEO-turnover difference-in-means table uses **220 second or later rounds**, split into **180** without turnover and **40** with turnover.
- The proximity probits use between **548** and **700** investor–firm pairs depending on whether ownership stakes are observed.

Interpretation. Different tests use different subsamples because the paper requires detailed matching of financing, board, and investor-location information.

3.9 Alternative explanation addressed in the text

Hermalin and Weisbach (1988) propose a public-firm explanation in which insiders passed over for CEO leave the board, so firms then refill vacancies with outsiders.

Lerner argues this is unlikely in private venture-backed firms because voluntary departures by senior executives are rare: leaving often requires selling shares back to the firm at the same discounted price originally paid, which is a substantial financial penalty.

Interpretation. This does not fully rule out all alternatives, but it weakens an important non-monitoring explanation for the turnover result.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the main turnover result

The first source of identification is a *within-interval comparison*: board changes are compared across otherwise similar financing intervals that do or do not contain CEO turnover.

The second source is *type specificity*: if turnover simply caused generic board reshuffling, one would expect both venture capitalists and other outsiders to increase. Instead, the large effect appears only for venture capitalists.

Why this matters. The specificity result is the paper’s strongest argument that it is identifying something special about venture-capital monitoring.

4.2 Why reverse causality is weaker than usual

A concern could be that venture-capitalist board expansion itself causes CEO turnover rather than the reverse. The paper does not fully solve this timing problem, but the interpretation is helped by the fact that CEO replacements are plausibly responses to organizational stress and uncertainty, not routine retirements.

The median exiting CEO is only **40** years old, and only **one** turnover resembles ordinary retirement.

Interpretation. The evidence is more consistent with “firm trouble or transition triggers extra oversight” than with “venture capitalists randomly add board seats and then replace CEOs.”

4.3 Controls for financing growth and board churn

Two direct confounds are addressed in the Poisson regressions.

- **Financing growth:** larger funding rounds may bring in more investors, who naturally want seats.
- **Vacated seats:** when directors leave, new directors may mechanically replace them.

The paper controls for both the change in dollars invested and the number of departing board members.

Interpretation. The CEO-turnover coefficient is not just picking up bigger financing rounds or simple seat refilling.

4.4 Why the promotion-story alternative is less persuasive here

The public-firm story that passed-over insiders leave the board after a new CEO is chosen is less compelling in this setting because managers in private firms often face a heavy penalty if they leave voluntarily.

Interpretation. The paper argues that voluntary exits of disappointed insiders should be uncommon, so the observed outsider changes are less likely to be a mechanical consequence of insider departures.

4.5 Why the proximity evidence supports costly monitoring

The proximity result is important because it comes from a different margin than CEO turnover. If venture-capital board service were purely symbolic certification, one might expect less sensitivity to distance. Instead, after controlling for stake and venture-organization characteristics, more distant investors are significantly less likely to hold board seats.

Interpretation. This cross-sectional pattern fits the idea that oversight involves real time, travel, and organizational effort.

4.6 Sample quality and measurement credibility

The paper spends substantial effort validating the financing sample and cleaning venture rounds.

- sample coverage is high relative to patents, industry events, and financing,
- financing amounts are approximately unbiased,

- and board information is assembled from multiple independent sources.

Interpretation. Measurement error surely remains, but it is unlikely that the paper’s results are a simple artifact of a noisy commercial database.

4.7 What the paper can and cannot distinguish

The paper gives evidence consistent with monitoring, but it cannot completely separate:

- monitoring from certification,
- formal board oversight from broader venture “value added,”
- local monitoring-cost effects from local network and information advantages,
- and causal turnover effects from all possible omitted crisis-related shocks.

It also does not directly observe the informal visits, customer meetings, and strategic advice that venture capitalists may provide.

Interpretation. The evidence is strong for an *active governance role*, but it is not a complete structural decomposition of why venture capitalists add value.

4.8 Relation to the literature

The paper complements several strands of research:

- Barry et al. (1990) on venture involvement in going-public firms,
- Gompers (1994) and Sahlman (1990) on how venture capital addresses agency problems,
- and the banking literature on inside financiers and relationship lending, such as James (1987), Petersen and Rajan (1994), and Kaplan and Minton (1994).

Interpretation. Lerner’s contribution is to provide direct empirical evidence that venture-capital involvement on boards behaves the way a monitoring technology should behave.

5 Tables and figures

5.1 Table I: the corrected financings sample

Read.

- The final corrected sample contains **653 financing rounds** for **271 biotech firms** between 1978 and 1989.
- Financing activity is concentrated in the second half of the 1980s.
- Later rounds are much larger than early rounds: the mean round size increases from **\$2.11 million** in first rounds to **\$3.94 million** in second rounds and **\$6.14 million** in third rounds.

Economic message. The firms in the sample fit the classic staged-financing model of venture capital, with repeated rounds and larger later investments.

Table I

The Corrected Financings Sample

The table presents the number of financing rounds of private biotechnology firms in the corrected Venture Economics sample, the total dollars disbursed, and the average size^a of each round (in millions of 1989 dollars). The sample consists of 653 financing rounds of 271 biotechnology firms between 1978 and 1989. Financing rounds are segmented by year and by round number.

Panel A. Financings Segmented by Year			
Year	Number of Rounds	Aggregate Size (1989 \$ Millions)	Average Size (1989 \$ Millions)
1978	7	17.23	2.46
1979	8	54.43	6.80
1980	16	123.45	7.72
1981	41	140.88	3.61
1982	46	195.17	4.34
1983	62	227.88	3.80
1984	46	142.84	3.32
1985	55	149.14	3.04
1986	79	269.94	3.70
1987	102	359.98	3.79
1988	102	307.30	3.23
1989	89	336.46	4.31

Panel B. Financings Segmented by Round Number			
Financing Round	Number of Rounds	Aggregate Size (1989 \$ Millions)	Average Size (1989 \$ Millions)
First round	270	527.10	2.11
Second round	186	689.31	3.94
Third round	113	651.02	6.14
Later round	84	457.28	5.94

^a Because I cannot determine the size of some financing rounds, the aggregate size does not equal in all cases the product of the number of rounds and the average round size.

Table II

The Board Membership of Private Biotechnology Firms

The sample consists of 653 financing rounds of 271 biotechnology firms between 1978 and 1989; I present the board membership by round for each of the 362 rounds where membership can be determined. Venture Capitalists are defined as individuals who are general partners or associates at venture capital organizations that are either unaffiliated with any other organization or else affiliated with a financial institution. I count full-time affiliates of a venture capital organization as venture capitalists, even if they work for a venture-backed firm. Other Outsiders include corporate investors, other investors (individuals who (i), either alone or in a partnership, held a five percent stake in the organization at some time, (ii) never were an officer of the firm, and (iii) never were an affiliate of a company which signed a collaborative arrangement with the firm or of a venture investor), and individuals that do not have another relationship with the firm. Insiders are either senior (the chief executive officer, president, and chairman of the board) or junior managers employed directly by the firm. Quasi Insiders are those parties who do not work directly for the firm, but who have an ongoing relationship with the concern. Panel B reports the professional affiliation of board members at the time of the last financing round in the sample.

Panel A. Board Membership by Round Number				
Financing Round	Mean Number of Board Members			
	Venture Capitalists	Other Outsiders	Insiders	Quasi Insiders
First round	1.40	0.86	1.28	0.52
Second round	1.87	0.86	1.40	0.56
Third round	2.09	1.02	1.61	0.67
Later round	2.12	1.27	1.73	0.54

Panel B. Professional Affiliation of Board Members at Time of Last Financing Round (%)	
Outside directors	
Venture capitalist	36.2
Corporate partner	6.4
Other investor	3.1
Executive with other health care or biotechnology firm	3.5
Retired health care or high-technology executive	3.6
Academic without firm affiliation	0.9
Lawyer, consultant, or investment banker without firm affiliation	1.4
Other or unidentified	5.1
Inside directors	
Senior manager	20.3
Junior manager	7.1
Quasi-inside directors	
Academic affiliated with the firm	8.9
Lawyer affiliated with the firm	0.5
Investment or commercial banker affiliated with the firm	1.0
Former manager of the firm	0.6
Relative or other	1.3

5.2 Table II: board membership of private biotechnology firms

Read.

- Mean board size rises from about **4.06** in the first round to about **5.66** in later rounds.
- Venture capitalists hold **1.40** seats in the first round and **2.12** seats in later rounds.
- At the last financing round, venture capitalists account for **36.2%** of all board seats, more than any other identifiable group.
- Senior managers account for **20.3%** of seats and junior managers for **7.1%**.

Economic message. Venture capitalists are deeply embedded in formal governance of private venture-backed biotech firms, not peripheral advisers.

5.3 Table III: board changes around CEO turnover

Read.

- In **180** second-or-later rounds without CEO turnover, firms add only **0.24** venture directors on average.
- In **40** rounds with CEO turnover, firms add **1.75** venture directors.
- The increase in other outsiders is tiny by comparison: **0.28** without turnover and **0.33** with turnover.
- The difference for venture directors is highly significant in both t-tests and Wilcoxon tests; the outsider differences are not.

Economic message. When monitoring needs rise, the marginal board seat goes specifically to a venture capitalist.

Table III
The Changes in Board Membership between Financing Rounds

The sample consists of 220 second or later financing rounds where the board membership at the time of the current and previous round can be determined. Panel A indicates the change in board membership since the last financing round, divided by whether chief executive officer (CEO) turnover occurred. Venture Capitalists are defined as individuals who are general partners or associates at venture capital organizations that are either unaffiliated with any other organization or else affiliated with a financial institution. I count full-time affiliates of a venture capital organization as venture capitalists, even if they work for a venture-backed firm. Other Outsiders include corporate investors, other investors (individuals who (i), either alone or in a partnership, held a 5 percent stake in the organization at some time, (ii) never were an officer of the firm, and (iii) never were an affiliate of a company that signed a collaborative arrangement with the firm or of a venture investor), and individuals that do not have another relationship with the firm. Insiders are managers employed directly by the firm. Quasi Insiders are those parties who do not work directly for the firm, but who have an ongoing relationship with the concern. Panel B presents *p*-values from *t*-tests and non-parametric Wilcoxon tests of whether the change in board membership differs in rounds with CEO turnover. The *t*-tests do not assume that the two distributions have the same variance.

Panel A. Changes in Board Membership between Financing Rounds				
	Mean Change in the Number of Board Members Since the Last Financing Round			
	Venture Capitalists	Other Outsiders	Insiders	Quasi Insiders
180 rounds without CEO turnover	+0.24	+0.28	+0.10	+0.06
40 rounds with CEO turnover	+1.75	+0.33	+0.23	+0.25
Panel B. Tests of Equality of Changes in Board Membership between Financing Rounds				
	<i>p</i> -Value, Test of Null Hypothesis of No Difference between CEO Turnover Rounds and Other Rounds			
	Venture Capitalists	Other Outsiders	Insiders	Quasi Insiders
<i>p</i> -value, <i>t</i> -test	0.000	0.750	0.339	0.140
<i>p</i> -value, Wilcoxon test	0.000	0.519	0.094	0.297

Table IV
Poisson Regression Analysis of the Addition of Board Members between Financing Rounds

The sample consists of 216 second or later financing rounds where the board membership at the time of and the amount invested in the current and previous round can be determined. In Panel A, I estimate separate regressions using the number of new directors who are venture capitalists and other outsiders as the dependent variable. Independent variables include a dummy indicating if there was chief executive officer (CEO) turnover between the previous and current round, the difference in the amount invested in the current and previous round (expressed in millions of 1989 dollars), and the number of board members who departed the board between the previous and current round. Absolute *t*-statistics are in brackets. In Panel B, I test whether the coefficients of the CEO turnover variable in the two regressions are equal.

Panel A. Poisson Regression Analysis of the Addition of Board Members between Financing Rounds		
	Dependent Variable: Number of New Board Members who are...	
	Venture Capitalists	Other Outsiders
CEO turnover	1.88 [8.86]	-0.04 [0.13]
Change in dollars invested	0.02 [0.89]	-0.06 [2.28]
Number of departing board members	0.15 [2.06]	0.19 [1.41]
Constant	-1.43 [8.94]	-1.13 [8.00]
Log likelihood	-175.29	-157.20
χ^2 -statistic	112.50	7.56
<i>p</i> -Value	0.000	0.056
Number of observations	216	216
Panel B. Test of the Equality of Coefficients in the Venture Capitalist and Other Outsider Regressions		
<i>p</i> -Value, χ^2 -test of null hypothesis that CEO turnover coefficients are equal	0.000	

5.4 Table IV: Poisson regressions for adding board members

Read.

- The CEO-turnover coefficient is **1.88 [8.86]** in the venture-capitalist count regression.
- At the mean of the controls, CEO turnover raises expected new venture directors from about **0.25** to **1.59**.
- In the other-outsider regression, the CEO-turnover coefficient is **-0.04 [0.13]**, economically and statistically negligible.
- The equality of coefficients across the two regressions is rejected with p-value **0.000**.

Economic message. The turnover result survives controls and is sharply targeted at venture capitalists rather than outsiders in general.

5.5 Table V: proximity and board membership

Table V
The Relationship between Proximity and Board Membership for Venture Investors

I first present the distance (in miles) between the headquarters of the firm and the nearest office of each venture capitalist that served as a board member at the time of the last venture round in the sample. The sample consists of 700 pairs of venture capital organizations and private biotechnology firms. Panel B presents the relationship between probability of a venture investor serving as a board member and its distance to the firm, and the p -value from an F -test of this pattern. Panel C presents a probit regression analysis of the relationship between venture investor proximity and board membership. The dependent variable is a dummy indicating whether a representative of the venture organization served on the board at the time of the last venture round in the sample. (1.0 denotes a director.) Independent variables include the distance from the venture organization's nearest office to the headquarters of the firm (in thousands of miles), the fraction of the firm's equity held by the venture organization, the age of the venture organization (in years), and its size (the ratio of its committed capital to the total venture capital pool). I calculate all variables at the time of the last venture round in the sample. Absolute t -statistics are in brackets.

Panel A. Proximity of Venture Capitalist Directors				
	Distance from Venture Capitalist's Nearest Office to Firm Headquarters (in Miles)			
	Mean	Median	First Quartile	Third Quartile
Nearest venture director	359	59	7	418
Median distance venture director	584	287	32	965
Furthest venture director	993	419	73	1951

Table V—Continued

Panel B. Relationship between Proximity of Venture Investor and Probability of Board Membership				
	Distance from Venture Capitalist's Nearest Office to Firm Headquarters (in Miles)			
	< 5	5–50	50–500	> 500
Probability of joining board (%)	46.7	30.7	34.9	21.8
p -Value, F -test of null hypothesis of no relationship				0.000

Panel C. Regression Analysis of Board Membership				
	Dependent Variable: Venture Investor Served on Board			
	Using Venture Capitalist Age and Stake	Using Venture Capitalist Size and Stake	Using Venture Capitalist Age	Using Venture Capitalist Size
Venture office to firm (000 miles)	-0.18 [3.72]	-0.20 [4.04]	-0.16 [4.16]	-0.20 [4.71]
Stake held by venture organization	4.21 [6.96]	4.19 [6.61]		
Age of venture organization (years)	0.01 [1.54]		0.01 [2.18]	
Organization's share of total venture pool		18.64 [2.17]		18.62 [2.77]
Constant	-0.86 [7.33]	-0.87 [7.57]	-0.51 [5.80]	-0.48 [6.07]
Log likelihood	-319.48	-297.36	-413.19	-384.85
χ^2 -statistic	82.68	85.68	27.12	35.66
p -Value	0.000	0.000	0.000	0.000
Number of observations	580	548	700	661

Read.

- The nearest venture director is close for many firms: mean **359 miles**, median **59 miles**, first quartile **7 miles**.
- More than half the firms have a venture director within **60 miles**; one-quarter have one within **7 miles**.
- A venture investor whose nearest office is within **5 miles** serves on the board with probability **46.7%**; if the nearest office is more than **500 miles** away, the probability is only **21.8%**.
- In probit regressions, the distance coefficient lies between about **-0.16** and **-0.20**, while the ownership-stake coefficient is around **4.2** and strongly positive.

Economic message. Board oversight by venture capitalists is geographically local, consistent with monitoring being costly and contact-intensive.

6 Short summary

- The paper studies whether venture capitalists behave like active monitors of private firms.
- It uses a corrected panel of **271 venture-backed biotech firms** and **653 financing rounds** from 1978 to 1989.
- Venture-capitalist board representation rises sharply around **CEO turnover**, while other outsiders do not show the same pattern.
- Venture-capitalist board service is strongly related to **geographic proximity** to the firm, even after controlling for stake and venture-capitalist characteristics.
- The evidence supports the view that venture capital provides governance and monitoring, not just money.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes two central facts. First, venture capitalists add board seats when oversight needs are plausibly higher, as proxied by CEO turnover. Second, venture capitalists are more likely to sit on the boards of nearby firms, consistent with the idea that oversight is costly to provide.

7.2 Economic intuition

A venture capitalist is an informed inside investor. If monitoring adds value, it should appear where it is most useful and where it is cheapest. CEO succession is one moment when firms need particularly intense oversight. Geographic proximity lowers the cost of delivering that oversight through visits, meetings, and hands-on involvement. The empirical results line up with both implications.

7.3 Takeaway

Lerner's main message is simple and important: venture capitalists are not merely financiers who occasionally hold board titles. Their board roles behave like a real governance technology. That helps explain why venture capital can support young, intangible-asset-intensive firms that other financiers struggle to fund.